

Cover crop roots redesign the pore structure and organic matter composition to alter hydraulic properties of the rhizosphere

Renjie Ruan^{a,b}, Wanyu Xie^c, Zhongbin Zhang^{a,c,1,*}, Paul Hallett^d, Peng Xiong^e, Shuai Liu^c, Xinhua Peng^a

^a State Key Laboratory of Efficient Utilization of Arid and Semi-arid Arable Land in Northern China, Institute of Agricultural Resources and Regional Planning, Chinese Academy of Agricultural Sciences, Beijing 100081, China

^b College of Agronomy, Anhui Agricultural University, Hefei 230001, China

^c State Key Laboratory of Soil and Sustainable Agriculture, Institute of Soil Science, Chinese Academy of Sciences, Nanjing 210008, China

^d School of Biological Sciences, University of Aberdeen, Aberdeen AB24 3FX, UK

^e Key Laboratory of Agricultural Resources and Ecology in Poyang Lake Watershed of Ministry of Agriculture and Rural Affairs in China, College of Land Resource and Environment, Jiangxi Agricultural University, Nanchang 330045, China

ARTICLE INFO

This manuscript was handled by Amilcare Porporato, Editor-in-Chief, with the assistance of Salvatore Calabrese, Associate Editor

Keywords:

Rhizosphere hydraulic properties
Soil pore structure
Soil organic matter composition
Cover crops

ABSTRACT

Cover crop roots can considerably alter soil pore structure and organic matter composition in the rhizosphere, yet how these root-induced variations, which depend on cover crop species, influence hydraulic properties in the rhizosphere under field conditions remains poorly quantified. We integrated X-ray computed tomography (CT), Fourier-transform infrared (FTIR) spectroscopy, and mini-infiltrometer to characterize rhizosphere pore structure, organic matter composition, and hydraulic properties of three cover crops' tap roots (alfalfa, rapeseed, vetch) in a field Vertisol. The rapeseed exhibited enhanced rhizosphere macroporosity likely due to utilization of pre-existing macropores. Conversely, the alfalfa and vetch roots increased total carbon (C) content and potential wettability index (PWI) of the rhizosphere. All the cover crops reduced rhizosphere matrix density due to organic matter enrichment. The alfalfa and vetch rhizosphere developed significant water repellency, driving decreased sorptivity in rhizosphere. PWI-mediated water repellency explained most of the sorptivity variation, and matrix density-organic matter interactions additionally regulated rhizosphere sorptivity, independent of macroporosity. Our findings demonstrate distinct rhizosphere engineering strategies: the rapeseed facilitates rhizosphere-water sorptivity, accompanied with a high-macroporosity, low-matrix density, and minimal hydrophobicity rhizosphere; whereas the alfalfa and vetch suppress water sorptivity via organic matter controls on wettability, thereby slowing the rapid drying of the rhizosphere. Under field conditions, rhizosphere hydraulics are regulated by the combined effect of soil organic matter composition and matrix density, thereby revealing new patterns in plant-soil-water interactions.

1. Introduction

The hydraulic properties of the rhizosphere critically regulate root water uptake, shaping plant drought resilience and resource acquisition (Carminati et al., 2016; Whalley et al., 2005). Plant species vary significantly in their capacity to modify rhizosphere hydraulics through root penetration, expansion, and exudate-mediated interactions with the soil matrix (Chen and Weil, 2010; Rabbi et al., 2018; Zhang and Peng, 2021). For example, early studies revealed that barley roots release

hydrophobic exudates enhancing rhizosphere water repellency (Hallett et al., 2003), while drought-tolerant chickpea forms a more porous rhizosphere to improve water sorptivity (Rabbi et al., 2018). Although these studies have advanced our understanding of organic-matter and pore-structural drivers of rhizosphere hydraulics (Carminati et al., 2013; Leue et al., 2015), the synergistic interplay between soil structural reorganization and organic matter composition, specifically how plant species regulate these processes remains poorly quantified and mechanistically unresolved.

* Corresponding author at: State Key Laboratory of Efficient Utilization of Arid and Semi-arid Arable Land in Northern China, Institute of Agricultural Resources and Regional Planning, Chinese Academy of Agricultural Sciences, Beijing 100081, China.

E-mail address: zhangzhongbin@caas.cn (Z. Zhang).

¹ 0000-0003-0420-8063.

Soil pore structure fundamentally governs hydraulic processes by mediating water transport pathways and connectivity (Daly et al., 2015; Rabbi et al., 2018). Root growth induces radial stresses that reorganize soil particles, creating spatially heterogeneous pore networks characterized by localized densification and cracking (Aravena et al., 2011; Dexter, 1987; Huang et al., 2022). X-ray CT has enabled non-invasive characterization of these dynamic modifications, revealing seemingly contradictory patterns: while wheat rhizosphere exhibit reduced macroporosity (Daly et al., 2015), species like tomato and pea increase macroporosity through rhizosphere cracking (Helliwell et al., 2017; Helliwell et al., 2019). This divergence likely reflects species-specific root-soil interactions and contextual soil mechanics (Burr-Hersey et al., 2020; Koebnick et al., 2019; Koebnick et al., 2017). The hydraulic consequences are also paradoxical: while rhizosphere densification may enhance capillary continuity through improved particle contacts (Aravena et al., 2013), preferential macropores in rhizosphere can conversely disrupt matrix flow pathways (Helliwell et al., 2019). How plants strategically regulate these opposing processes to optimize water acquisition under field conditions represents a critical knowledge gap (Berli et al., 2008; Carminati et al., 2013).

The composition of soil organic matter constitutes another critical determinant of rhizosphere hydraulics (Gregory, 2006; Leue et al., 2015). Root-sourced biochemical inputs, including exudates and residues, alter soil organic matter composition and consequent wettability dynamics (Leue et al., 2010). A key mechanism is the accumulation of hydrophobic compounds, which can render the rhizosphere less wettable than the bulk soil (Leue et al., 2013). This feature of soil organic matter can be analysed using FTIR, including carboxyl (C=O) groups that are responsible for hydrophilicity or alkyl (C-H) groups that account for hydrophobicity (Heller et al., 2015). Leue et al. (2013) employed FTIR to reveal that the relative abundance of alkyl (C-H) to carboxyl (C=O) functional groups governs the soil PWI. The spatial variations in soil PWI and organic matter composition around plant roots subsequently determine water repellency in the rhizosphere (Bengough, 2012; Leue et al., 2010). Thus, the barley rhizosphere exhibited lower water sorptivity than the grass rhizosphere, due to the enrichment of hydrophobic compounds (Hallett et al., 2003), underscoring the species-specific influence of soil organic matter on hydraulic properties (Rabbi et al., 2018).

Emerging consensus confirms that rhizosphere pore structure or organic matter composition drive hydraulic modifications with specific ecological benefits, such as improving soil structural resilience and plant water availability (Naveed et al., 2017; Rabbi et al., 2018). However, critical knowledge gaps persist, as existing studies analyze rhizosphere pore structure or organic matter composition in isolation, predominantly using repacked soils under controlled greenhouse conditions (Burr-Hersey et al., 2020; Lucas et al., 2022). This methodological deficiency creates systematic bias, as laboratory experiments consistently overestimate the magnitudes of rhizosphere hydraulic modifications compared to field studies (Hallett et al., 2003). In addition, current evidence remains narrowly focused on cereal crops (e.g., wheat, maize), despite their limited root system plasticity (Zheng et al., 2023). Cover crops, specifically selected for rhizosphere engineering due to their high plasticity to soil strength, present a promising avenue for soil hydraulic reconfiguration (Zhang et al., 2022). Research has suggested several mechanisms by which cover crops might contribute: potential generation of biomechanical forces by thick taproots (e.g., rapeseed) to create persistent macropore networks; potential preferential colonization of pre-existing macropores (e.g., root channels, biopores) to modulate porosity; and potential establishment of biochemical gradients through hydrophobic exudates released by dense root hairs to regulate preferential flow (Blanco-Canqui and Ruis, 2020; Lucas et al., 2022; Zhang et al., 2022). Critically, field conditions likely promote interactions between these mechanisms in cover crops (Lucas et al., 2022), while the underlying effects of soil pore structure and organic matter composition on rhizosphere hydraulics, particularly their in situ synergistic

regulation mechanisms remain largely speculative.

This study specifically examines taproots of alfalfa, rapeseed, and vetch due to their great potential in rhizosphere engineering (Zhang et al., 2022). Although lateral roots are primary sites for water uptake, taproots serve as critical transport conduits and generate radial stresses that reshape macropore networks: key determinants of hydraulic connectivity in compacted Vertisols (Zarebanadkouki et al., 2013). As most studies on rhizosphere pore structure have used filled soil columns with homogeneous structure, neglecting field soil with heterogeneous structure, we sought a mechanistic understanding of the root-soil interface using a field sampled soil cores by (1) measuring soil macroporosity and matrix density near the root surface using X-ray CT, and (2) assessing soil organic matter composition in the rhizosphere and bulk soil using FTIR spectroscopy. Additionally, we measured water sorptivity, ethanol sorptivity, and water repellency in both rhizosphere and bulk soil at small-scale using a mini-infiltrometer. The integrated application of X-ray CT, FTIR spectroscopy, and mini-infiltrometry enables a novel multi-scale assessment, linking rhizosphere structural and chemical properties directly to hydraulic functions under field conditions. We hypothesize that the root systems of these cover crops induce distinct changes in rhizosphere macroporosity, matrix density, and organic matter content, synergistically influencing water sorptivity and retention. This mechanistic framework offers new insights into the hydraulic dynamics of the rhizosphere under field conditions.

2. Materials and methods

2.1. Field site and experiment design

The experiment had a continuous winter cover crop-summer maize rotation and was established at Longkang Farm, Anhui Province, China (33°32' N, 115°59' E, and 23 m in elevation) in November 2017. The annual precipitation was 912 mm, and the mean annual temperature was 14.8°C at the experiment site, with a recorded precipitation of 306 mm and an average temperature of 11°C during the cover crop growing period from October to May. The soil was classified as a Vertisol with a soil texture of clay loam (sand, 8%; silt, 54.1%; and clay, 37.9%) (Soil Survey Staff, 2015).

The field experiment was designed with three replicates of three cover crop treatments: alfalfa, rapeseed, and a mixture of hairy vetch and radish. The field was tilled with a rotary tiller to a depth of 5–10 cm to prepare an appropriate seedbed for all the treatments. After maize harvest (in October), cover crops were planted by broadcasting, with seeding rates of 21.4 kg ha⁻¹ for the rapeseed and alfalfa and 42.8 kg ha⁻¹ for the mixture of hairy vetch and radish (1:1). The cover crops were harvested, and the aboveground residues were removed in the following May before maize sowing. Then, rotary tillage was performed to a depth of 5–10 cm and the basal fertilizers were applied (100 kg N ha⁻¹, 60 kg P₂O₅ ha⁻¹, and 90 kg K₂O ha⁻¹). Maize seeds were sown manually in all cover crop treatments, and topdressing fertilizer (80 kg N ha⁻¹) was applied for maize growth. Detailed field management was described in the study of Zhang et al. (2022).

2.2. Soil sampling and measurements

In May 2019 (mature stage for rapeseed, late flowering stage for alfalfa and vetch), intact soil cores with cover crop tap roots were collected vertically using PVC cylinders (25 mm height × 25 mm diameter) below the depth of rotary tillage at 10–20 cm in each plot. For each cover crop treatment, 10 cores were sampled per 3 replicated plots. Then, the soil cores were wrapped with plastic film and transported to the laboratory to avoid any disturbance. The soil cores were stored in a refrigerator at 4°C for the following measurements. The whole root of each cover crop is presented in Fig. S1, which is collected after the cover crops were harvested. The tap roots of alfalfa, rapeseed, and vetch with a diameter > 1 mm are preferred, avoiding lateral roots to isolate taproot-

specific effects.

2.3. X-ray scanning and image analysis

An industrial X-ray CT (Phoenix Nanotom; GE, Sensing and Inspection Technologies, GmbH, Wunstorf, Germany) was used to scan the soil cores with a voltage of 110 kV and a current of 110 μ A. The Datos|x 2.0 software and the filtered back-projection algorithm were used for image reconstruction. After reconstruction, approximately 2400 slices with a voxel size of $12 \times 12 \times 12 \mu\text{m}^3$ were generated. Ten soil cores were scanned for each cover crop treatment.

Image processing was performed using VG Studio Max software (Volume Graphics GmbH, Heidelberg, Germany). A region of interest (ROI) of 16.97 mm in diameter and 12.00 mm height was selected from the central part of each soil core to remove the edge effect caused by sampling. The procedures of root, pore, and matrix segmentation are depicted in Fig. S2. First, the image grayscale for all the soil cores was calibrated using the pure air space around the core samples (gray-value 50) and the PVC cylinder as the reference objects (gray-value 200). For root segmentation, an adaptive region growing algorithm was used within the user defined gray-value range (Helliwell et al., 2019). The images were also visually inspected to verify the quality of the root segmentation. Subsequently, the segmented root region was inverted to create spaces of pore and soil matrix. A gray-value range from 125 to 200 (beyond 50% of the calibrated range) was used to segment the soil matrix from the pore space (Helliwell et al., 2017). It comprised the following two regions: CT-derived pores (macropores region, 12 μm resolution) and CT-derived matrix (soil micropores region, solid soil phase + micropores with diameters less than 12 μm). The latter one is referred to as the soil matrix (Duddek et al., 2023).

The root region was three-dimensionally (3D) dilated with 18 voxels (216 μm side length) each time to a total of 180 voxels (2160 μm) dilations, creating 10 discreet regions moving away from the root surface, in which CT-derived macroporosity and matrix density were calculated (Fig. 1 and Fig. S2). The “Defect detection module (VG Studio Max software)” with default algorithm was used to determine the CT-derived macroporosity and pore size distribution within each dilated region. The soil pore size distribution was calculated with equivalent diameters of 12–100, 100–500, 500–1000, and > 1000 μm within each dilated

region. Furthermore, the grayscale intensity reflects the matrix density, which could be determined by analyzing the grayscale histograms without including the grayscale of the stones in the CT images (Katuwal et al., 2015). A gray-value range beyond 230 was used to segment the stones (Fig. S2). Then, the matrix density within each dilated region was calculated using the “Gray value detection module (VG Studio Max software)” with the default algorithm.

The root volume fraction of the whole ROI was calculated using the “Object properties tool” in VG Studio Max software. The root diameter was measured using the ImageJ software with the “Thickness in BoneJ plugins” (Zhang et al., 2018). The CT-derived macroporosity and matrix density at the locations of 0–1080 and > 2160 μm from the root surface represent these CT-derived soil characteristics of the rhizosphere and bulk soil, respectively.

2.4. Measurement of hydraulic properties

The soil cores after CT scanning were air dried and then carefully broken along the roots, which were removed subsequently. The locations at the root-soil interface and > 2 mm away from the interface, representing the rhizosphere and bulk soil, respectively, were selected to measure water and ethanol infiltration according to the method of Naveed et al. (2017) using a mini-infiltrometer with a tip radius of 0.35 mm. Two different locations were selected for the measurement in both the rhizosphere and bulk soil for each soil core. Ethanol infiltration provides a reference measurement not influenced by the hydrophobicity of the soil because of the nonpolar nature of ethanol and its contact angle with hydrophobic surfaces (Hallett et al., 2003). All measurements were conducted at a –10 mm hydraulic head. The weight of liquid taken up by the soil was logged using an electronic balance at 1 s intervals for 50 s (Fig. S3). The water sorptivity, ethanol sorptivity, and water repellency were calculated with the equations in Naveed et al. (2017).

2.5. Soil organic matter composition

We collected soil samples from air-dried cores using a knife. Soil samples were obtained from both the rhizosphere (within 1 mm from root-soil interface) and bulk soil near the location of hydraulic measurements, but outside of the zone of influence of water or ethanol

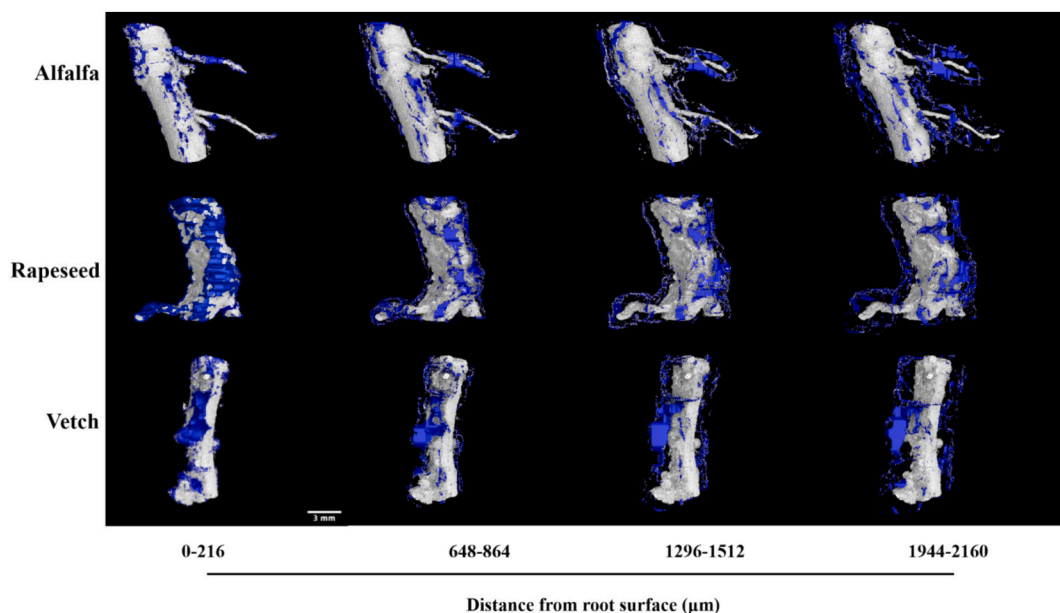


Fig. 1. Delineation of zones near root surface and examples of 3D identification and visualization of pore space in three cover crop treatments. The images were produced using VGStudio MAX v.2.2.5. White color represents the roots of cover crops. Blue color represents the soil pores in each distance from root surface. Samples were collected at 10–20 cm depth. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

infiltration. The soil total C and nitrogen (N) contents were measured using a CNS organic elemental analyser (Vario MAX, Elementar, Germany) to characterize soil organic matter content.

The soil organic matter composition was characterized using attenuated total reflectance (ATR)-FTIR spectroscopy (Nicolet 8700; Thermo Fisher Scientific, Waltham, MA, USA). Approximately 0.5 g of soil samples were spread onto the ATR-FTIR crystal. For each spectrum, 32 scans were compiled to give 1 spectrum ranging from 4000 to 650 cm^{-1} with a resolution of 4 cm^{-1} . The software OMNIC 9.1 (Thermo Nicolet, Waltham, MA, USA) was used to analyse the spectral data. The obtained FTIR spectra were analysed semi-quantitatively in regions of A_a (2966–2800 cm^{-1}), A_s (2880–2820 cm^{-1}), B_1 (1720–1700 cm^{-1}), and B_2 (1640–1600 cm^{-1}). Bands A_a and A_s correspond to the asymmetric and symmetric aliphatic C–H stretching, respectively (Verchot et al., 2011). The peak areas of bands A_a and A_s were quantified as the vertical distance from a local baseline plotted between tangential points, as these C–H stretch vibrations of alkyl groups typically appear as a shoulder of the broad OH band (Leue et al., 2010). Bands B_1 and B_2 correspond to the carboxylic-C (C=O stretching of carboxyl and ketones) and

aromatic-C (aromatic C=C conjugated with C=O and/or COO–) stretching, respectively (Leue et al., 2015; Verchot et al., 2011). The peak areas of bands B_1 and B_2 were measured as the height from the total baseline of the spectra. Following Leue et al. (2015), the C–H group (A_a and A_s) reflects the hydrophobic functionality, whereas the C=O group (B_1 and B_2) reflects the hydrophilic functionality. The PWI was calculated as the ratio of the summed C–H peak area to the summed C=O peak area, with larger PWI values indicating lower wettability (Leue et al., 2013).

2.6. Statistical analysis

All the obtained data were checked for normality and homogeneity of variance (Origin 2022). One-way analysis of variance was performed to assess the effect of cover crops and the differences among each distance from the root surface. Means comparisons were conducted using Fisher's least significant difference (LSD) at $P < 0.05$. An independent-sample t test was applied to assess the difference between the rhizosphere and bulk soil.

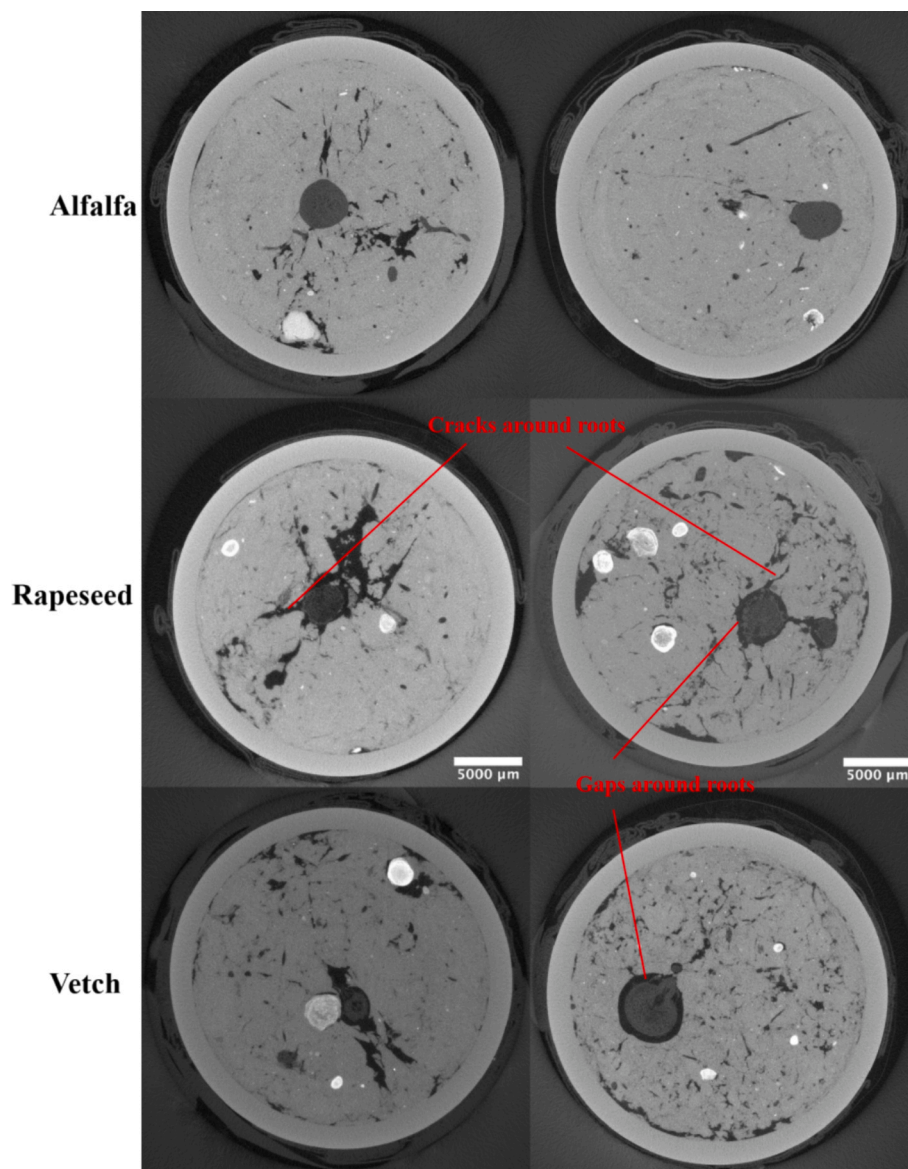


Fig. 2. Representative raw grayscale X-ray CT images showing cracks and gaps around three cover crop roots. Light gray represents soil matrix; moderate gray represented roots; dark gray represents pore space. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

We used linear regression analyses to assess the relationships among soil organic matter composition and hydraulic properties (Origin 2022). We further assessed the bivariate correlations between the CT-derived soil characteristics and rhizosphere hydraulic properties using zero-order correlations and partial correlations (Pearson correlation, SPSS 26) by controlling for the variables of total C and PWI. Also, the result calculated from a partial regression analysis was used to plot the relationship between matrix density and ethanol sorptivity.

3. Results

3.1. Rhizosphere pore structure

X-ray CT quantification revealed distinct spatial patterns of pore structure reorganization across cover crop treatments (Figs. 1 and 2). All treatments exhibited a radial gradient in CT-derived macroporosity, decreasing with distance from the root surface. This spatial decrease

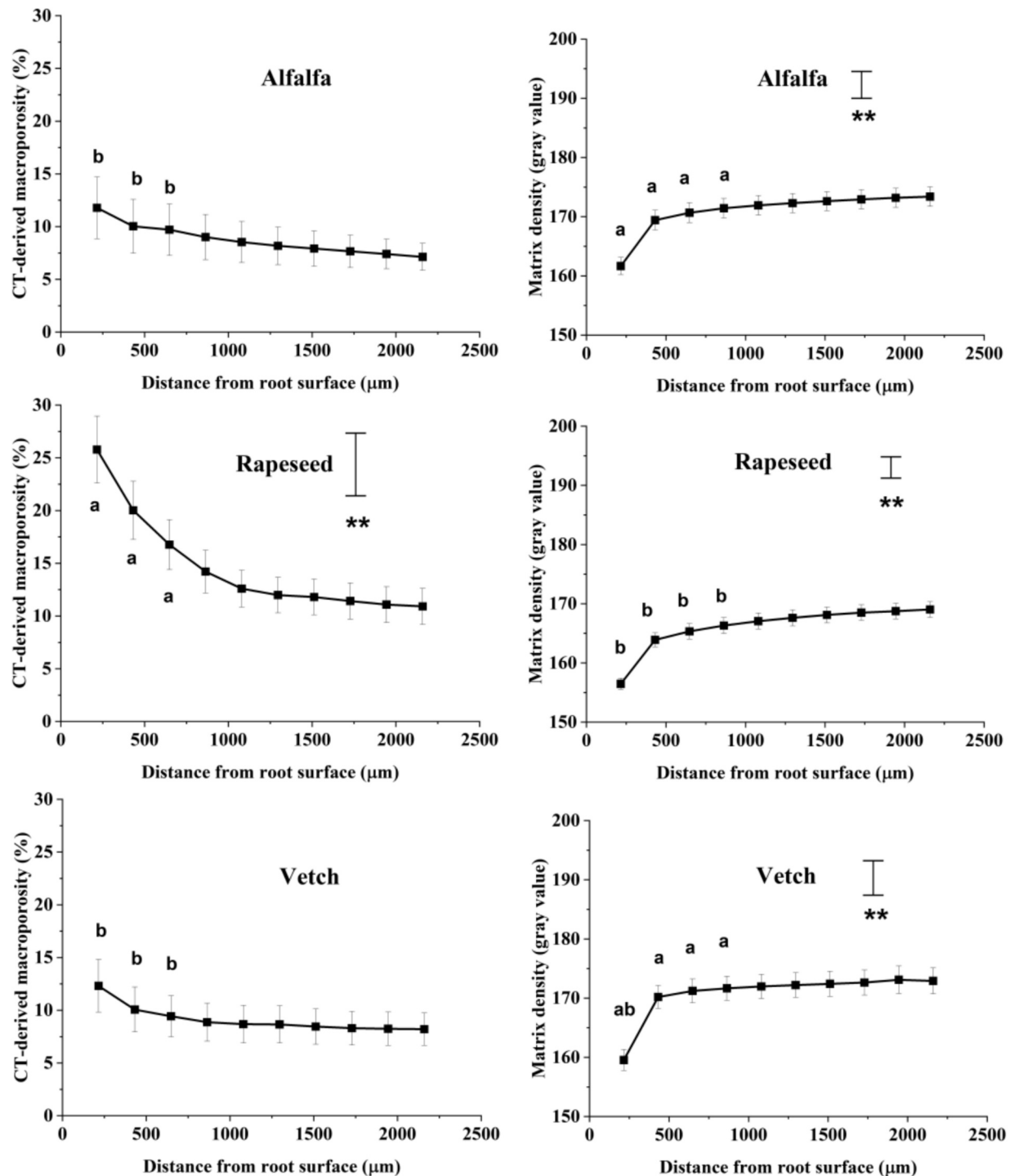


Fig. 3. CT-derived soil characteristics with distances from root surface in three cover crop treatments. Different letters indicate significant differences among three cover crop treatments within same distance from root surface at $P < 0.05$; stars indicate significant difference among the each distance in each cover crop treatment (*, $P < 0.05$; **, $P < 0.01$); and vertical bars indicate significant LSD values. Means and standard errors are presented (n = 10). Samples were collected at 10–20 cm depth.

showed strong cover crop species dependence: the rapeseed exhibited significant macroporosity reduction with radial distance (Fig. 3, $P < 0.01$), contrasting with non-significant trends in the alfalfa and vetch. Specifically, macroporosity within the immediate rhizosphere (distances 0–648 μm) induced by rapeseed roots was significantly higher ($P < 0.05$) than that induced by alfalfa or vetch. This spatial differentiation was particularly pronounced in pores $> 1000 \mu\text{m}$ diameter, where the rapeseed maintained significantly greater porosity than the alfalfa and vetch in the 0–864 μm zone (Fig. S4, $P < 0.01$). Visual evidence from CT imaging confirmed this structural differentiation, with the rapeseed roots surrounding by distinctive gaps and irregular cracks networks at

root-soil interfaces (Fig. 2), which were less noticeable around the alfalfa and vetch roots.

Accompanied by macroporosity changes, matrix density displayed an inverse radial pattern, increasing significantly from root surface to bulk soil across all treatments (Fig. 3, $P < 0.01$). The rapeseed rhizosphere maintained significantly lower matrix density than the alfalfa and vetch in the 216–864 μm zone ($P < 0.05$).

Notably, root morphological parameters (volume fraction and diameter) showed no significant differences among the three cover crops within the analyzed ROI (Fig. 4). However, at the rhizosphere scale (0–1080 μm), the rapeseed maintained significantly greater

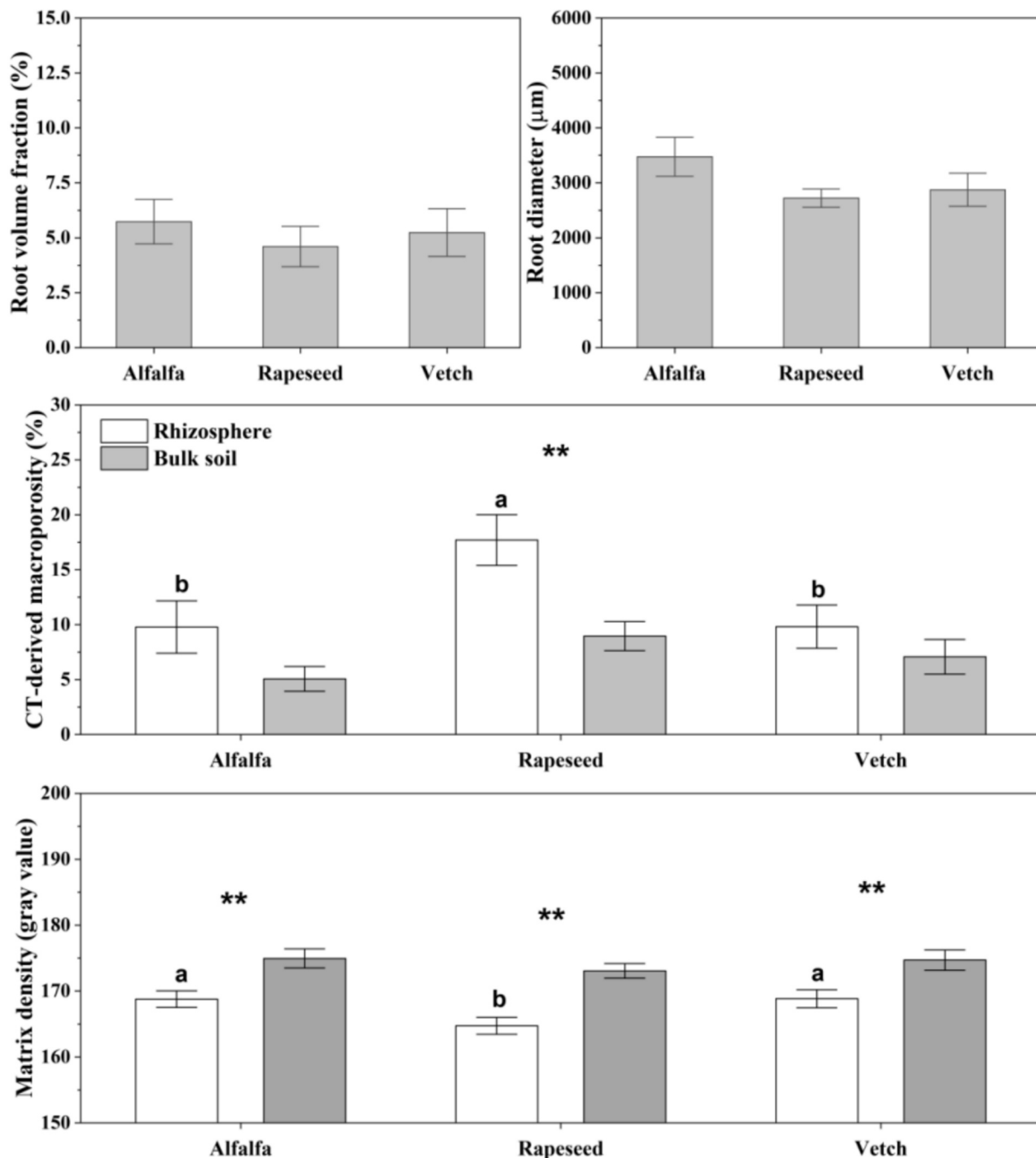


Fig. 4. The CT-derived characteristics of root, rhizosphere and bulk soil in three cover crop treatments. The locations with 0–1080 and $> 2160 \mu\text{m}$ from the interface represent the rhizosphere and bulk soil, respectively. Different letters indicate significant differences among three cover crop treatments at $P < 0.05$. Stars indicate significant difference between the rhizosphere and bulk soil in each cover crop treatment (**, $P < 0.01$). Means and standard errors are presented ($n = 10$). Samples were collected at 10–20 cm depth.

macroporosity and less matrix density than the alfalfa and vetch ($P < 0.05$). These structural modifications remained localized, as bulk soil ($> 2160 \mu\text{m}$) displayed no cover crop-specific differences in macroporosity or matrix density.

3.2. Rhizosphere organic matter composition and rhizosphere wettability index

Significant enrichment of total C occurred in the rhizosphere compared to bulk soil under the alfalfa (+30.5%) and vetch (+25.6%) treatments (Table 1, $P < 0.05$), suggesting higher rhizosphere soil organic matter content under the alfalfa and vetch treatments. The FTIR spectral analysis revealed cover crops-specific organic matter composition (Fig. 5 and Table S2), where the alfalfa rhizosphere uniquely enhanced asymmetric aliphatic-C (+30.0%), symmetric aliphatic-C (+37.5%), and carboxylic-C (+54.3%) compared to the bulk soil ($P < 0.05$). This alteration resulted in a significant increase in carboxylic-C retention by alfalfa, relative to rapeseed and vetch, in both the rhizosphere and bulk soil ($P < 0.05$). While the vetch similarly increased asymmetric aliphatic-C in the rhizosphere relative to the bulk soil (+34.3%, $P < 0.05$), its modification remained insufficient to drive statistically significant cover crop differentiation within the rhizosphere.

The observed compositional shifts led to differential accumulation patterns of hydrophobic versus hydrophilic functional groups (Table S2). The alfalfa and vetch rhizosphere accumulated 32.5% and 35.7% more hydrophobic C-H groups (including asymmetric and symmetric aliphatic-C) than their respective bulk soils, whereas hydrophilic C=O groups (including carboxylic and aromatic-C) showed modest increases of 8.4% and 2.4% respectively. These compositional shifts drove marked alterations in soil wettability (Fig. 5). The PWI showed consistent rhizosphere enhancement, with the alfalfa and vetch rhizosphere exhibiting significantly higher PWI than the bulk soil, respectively ($P < 0.05$). Notably, the vetch rhizosphere maintained a significantly higher PWI than the alfalfa ($P < 0.05$), establishing its least water-wettable character.

3.3. Rhizosphere hydraulic properties

Across cover crop treatments, hydraulic properties in the rhizosphere differed markedly from those in the bulk soil (Fig. 6). The alfalfa and vetch treatments significantly reduced rhizosphere water sorptivity compared to bulk soil ($P < 0.05$), whereas the rapeseed displayed no significant rhizosphere-bulk differences. Ethanol sorptivity showed a similar trend: the alfalfa rhizosphere exhibited lower ethanol sorptivity than the bulk soil ($P < 0.05$). Furthermore, the alfalfa and vetch rhizosphere had significantly higher repellency than bulk soil ($P < 0.01$).

Table 1

Total soil carbon and nitrogen of rhizosphere and bulk soil subjected to three cover crop treatments.

		Total C	Total N
		g kg ⁻¹	g kg ⁻¹
Rhizosphere	Alfalfa	14.99 (0.99)	1.86 (0.10)
	Rapeseed	15.49 (1.62)	2.11 (0.27)
	Vetch	14.71 (0.61)	2.08 (0.23)
Bulk soil	Alfalfa	11.48 (0.39) *	1.50 (0.08) *
	Rapeseed	12.74 (0.24)	1.91 (0.12)
	Vetch	11.71 (0.49) **	1.68 (0.16)

Different letters indicate significant differences among three cover crop treatments within the rhizosphere or bulk soil at $P < 0.05$; Stars indicate significant difference between the rhizosphere and bulk soil in each cover crop treatment (*, $P < 0.05$; **, $P < 0.01$). Means and standard errors are presented ($n = 5$). Samples were collected at 10–20 cm depth.

Conversely, the rapeseed rhizosphere repellency was similar to the bulk soil. The vetch rhizosphere exhibited significantly greater water repellency than the rapeseed and alfalfa ($P < 0.05$) and a corresponding reduction in water sorptivity compared to the rapeseed rhizosphere ($P < 0.05$).

3.4. Relationships among rhizosphere organic matter composition, pore structure and hydraulic properties

As depicted in Fig. 7, organic matter composition dominated hydraulic properties, with total C content explaining 66–74% of ethanol and water sorptivity reductions through negative correlations ($P < 0.05$). The PWI exhibited stronger control, accounting for 88% of water repellency variation (positive, $P < 0.01$) and 73% of water sorptivity changes (negative, $P < 0.05$).

Soil pore structure provided additional modulation in the rhizosphere (Fig. 8). While rhizosphere macroporosity showed no direct correlations with hydraulic properties, rhizosphere matrix density negatively influenced ethanol sorptivity (Fig. 8, $P < 0.01$), explaining 20.7% of variance when controlling for total C effect.

4. Discussion

4.1. Rhizosphere pore structure: cover crop root-induced modifications

Our X-ray CT analyses revealed a consistent enhancement in rhizosphere macroporosity across cover crop treatments compared to bulk soil (Figs. 3 and 4). This finding challenges the classical view of rhizosphere densification due to radial root expansion pressures (Aravena et al., 2011), while corroborating recent observations of macroporosity development near the root surface (Burr-Hersey et al., 2020). Mechanistically, repacked soil experiments attribute macroporosity formation to root-induced cracking caused by lateral mechanical stresses (Helliwell et al., 2017; Helliwell et al., 2019), or to shrinkage effects driven by root water uptake and soil moisture fluctuations (Carminati et al., 2013; Koebnick et al., 2018). However, in our field experiment, the soil is highly structured with different types of macropores such as root and earthworm induced biopores and shrinkage and swelling induced cracks, quite different from repacked soil column experiment with homogeneous organization of soil particles in laboratory systems. Thus, we identified large irregular cracks around cover crop roots, which most likely were pre-existing macropores as the living roots usually produce cylindrical channels or small fissures around the channels (Fig. 2). In this Vertisol with high clay content and bulk density, cover crop roots likely exploit these pre-existing macropores, to bypass the compacted soil with great penetration resistance and poor aeration (Atkinson et al., 2020; Zhang and Peng, 2021; Zhang et al., 2022), resulting in poor contact between roots and macropore walls and subsequently increased macroporosity in the rhizosphere (Xiong et al., 2022). The positive correlation between CT-derived macroporosity in rhizosphere and bulk soil (Fig. S6) likely supports that pre-existing macropores contribute significantly to the higher macroporosity observed near roots, which has not been clarified in laboratory studies using repacking soils previously. Notably, cover crop-specific differences in root behaviors were also evident in their influence on rhizosphere macroporosity (Figs. 3 and 4). The rapeseed exhibited superior macroporosity development, potentially due to two factors: (1) weaker root activity of rapeseed near maturity (Fig. S1), leading to root shrinkage, and (2) greater reliance of rapeseed roots in utilizing pre-existing macropores (Fig. 2) because of its limited capacity to create new pores under high bulk density conditions (Athmann et al., 2019). In contrast, alfalfa roots were demonstrated to have a strong biological drilling capacity (Fig. 2), actively creating new pores as found in other research (Song et al., 2021). These findings suggest that root penetration strategies and preferences for utilizing macropores are plant species dependent and play a critical role in rhizosphere macroporosity

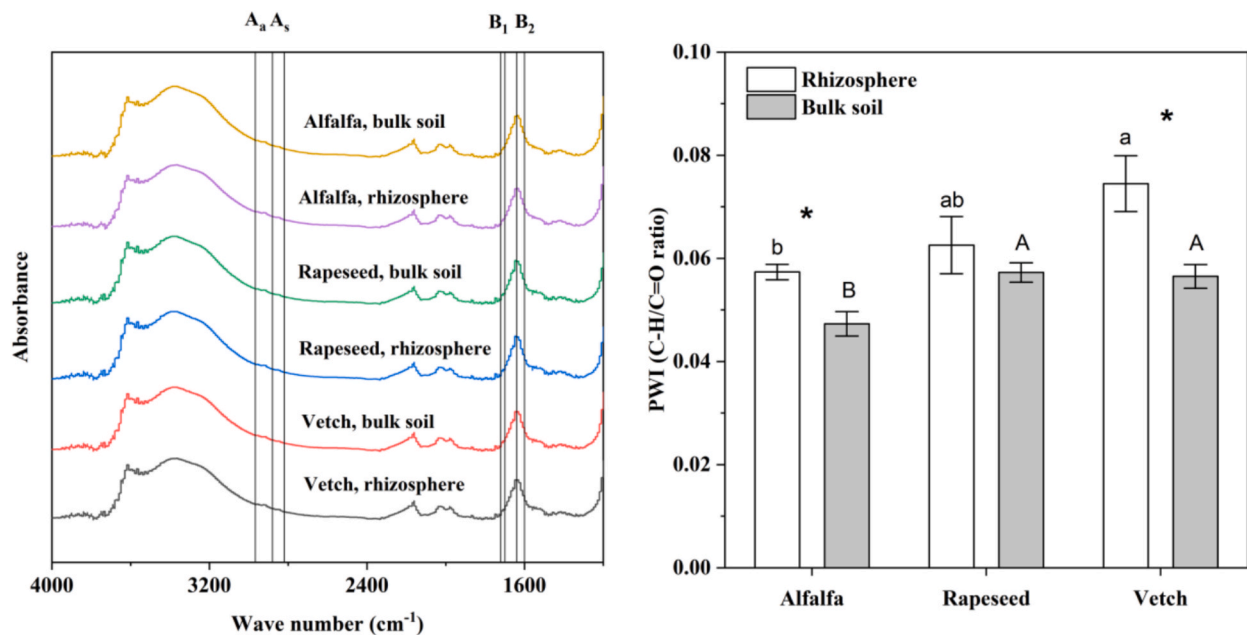


Fig. 5. Average FTIR spectra and potential wettability index (PWI) of rhizosphere and bulk soil in three cover crop treatments. A_a, asymmetric aliphatic C–H (2966–2880 cm⁻¹); A_s, symmetric aliphatic C–H (2880–2820 cm⁻¹); B₁, carboxylic C=O (1720–1700 cm⁻¹); B₂, aromatic C=O (1640–1600 cm⁻¹). Different letters indicate significant differences among three cover crop treatments within the rhizosphere or bulk soil at $P < 0.05$; Stars indicate significant difference between the rhizosphere and bulk soil in each cover crop treatment (*, $P < 0.05$; **, $P < 0.01$). Means and standard errors are presented ($n = 5$). Samples were collected at 10–20 cm depth.

formation under field conditions.

Concurrent matrix density reductions near roots across all the cover crop treatments indicate rhizosphere loosening (Figs. 3 and 4), given that matrix density correlates positively with bulk density (Katuwal et al., 2015). The reduced matrix density or bulk density in the rhizosphere probably resulted from higher organic matter content (Paradelo et al., 2016), which is supported by a negative correlation between matrix density and total C content in this study (Fig. S5). Soil organic matter derived from roots may contribute to reduced matrix density through two mechanisms: (1) enhancing elasticity between soil particles and increasing rebound capacity (Lin et al., 2022; Naveed et al., 2017); and (2) reducing overall density via the dilution effect of organic matter and soil aggregation (Blanco-Canqui et al., 2009; Lin et al., 2022). The findings highlight the capacity of cover crops to engineer rhizosphere matrix density through targeted organic enrichment.

4.2. Rhizosphere organic matter composition: drivers of wettability

Our FTIR analyses showed pronounced rhizosphere enrichment of total carbon (C) and aliphatic-C moieties, particularly under the alfalfa and vetch treatments (Tables 1 and Table S2). This spatial patterning aligns with numerous previous studies, showing that the rhizosphere exhibits elevated total C content relative to bulk soil (Leue et al., 2015; Lucas et al., 2023), primarily due to substantial inputs of root exudates and residues (Angst et al., 2016; Du et al., 2021). These organic residues undergo rapid degradation in the rhizosphere, forming labile components such as amino acids, which are detected in aliphatic-C signals in the FTIR spectra (Kunlanit et al., 2014). As highlighted by Du et al. (2021), the root-soil interface serves as a hotspot for the distribution of aliphatic-C derived from plant roots, thereby contributing to the observed increase in aliphatic-C content in the rhizosphere under the alfalfa and vetch treatments.

A significant difference was observed in the PWI (C–H/C=O ratio) between the alfalfa and vetch rhizosphere and the bulk soil (Fig. 5). This divergence resulted from differential stabilization pathways: hydrophobic aliphatic groups (C–H) were substantially increased in the

rhizosphere, whereas hydrophilic carboxylic and aromatic groups (C=O) showed limited enrichment. We attribute this compositional shift to the following processes: the carboxylic-C and aromatic-C fractions, including carbohydrates, degrade to produce aliphatic-C in the rhizosphere (Verchot et al., 2011). As reported in previous studies, this degradation intensifies when cover crops near maturity, leading to relatively greater increases in hydrophobic functional groups in the rhizosphere, and consequently, the greater PWI (Leue et al., 2013; Rumpel and Kögel-Knabner, 2011).

4.3. Hydraulic functionality: matrix density-organic matter interplay

Our results showed a positive relationship between the PWI and water repellency (Fig. 7), confirming organic matter composition as the principal regulator of rhizosphere hydrophobicity. Similar result was also reported by Leue et al. (2015), where PWI serves as a reliable predictor of root-induced water repellency. This phenomenon is commonly observed in soils with high organic matter, where organic matter composition greatly determines water repellency (Jordán et al., 2013). Similarly, the rhizosphere's role as an organic matter deposition hotspot creates self-reinforcing hydrophobicity through C–H group accumulation, which was particularly pronounced in alfalfa and vetch systems (Fig. 5). Consequently, the elevated PWI reduces water sorptivity (Fig. 7), potentially constraining the capacity of root water uptake in the rhizosphere (Villarreal et al., 2024).

Furthermore, the suppression of water and ethanol sorptivity through total carbon enrichment (Fig. 7) reveals a pore-clogging mechanism. As demonstrated by Hallett et al. (2003), root-derived organic matter can clog pores due to its swelling during wetting process. This is particularly evident in capillary networks, where organic matter generally fills soil micropores (Mikutta et al., 2004). Our findings extend this deduction by quantifying clogging severity: each 1% increase in rhizosphere C content reduced water sorptivity by 0.14 mm s^{-0.5} (Fig. 7), nearly identical to the 0.13 mm s^{-0.5} reduction per 1% exudate addition in repacked soils (Naveed et al., 2017). This convergence between field-based and laboratory-controlled systems establishes

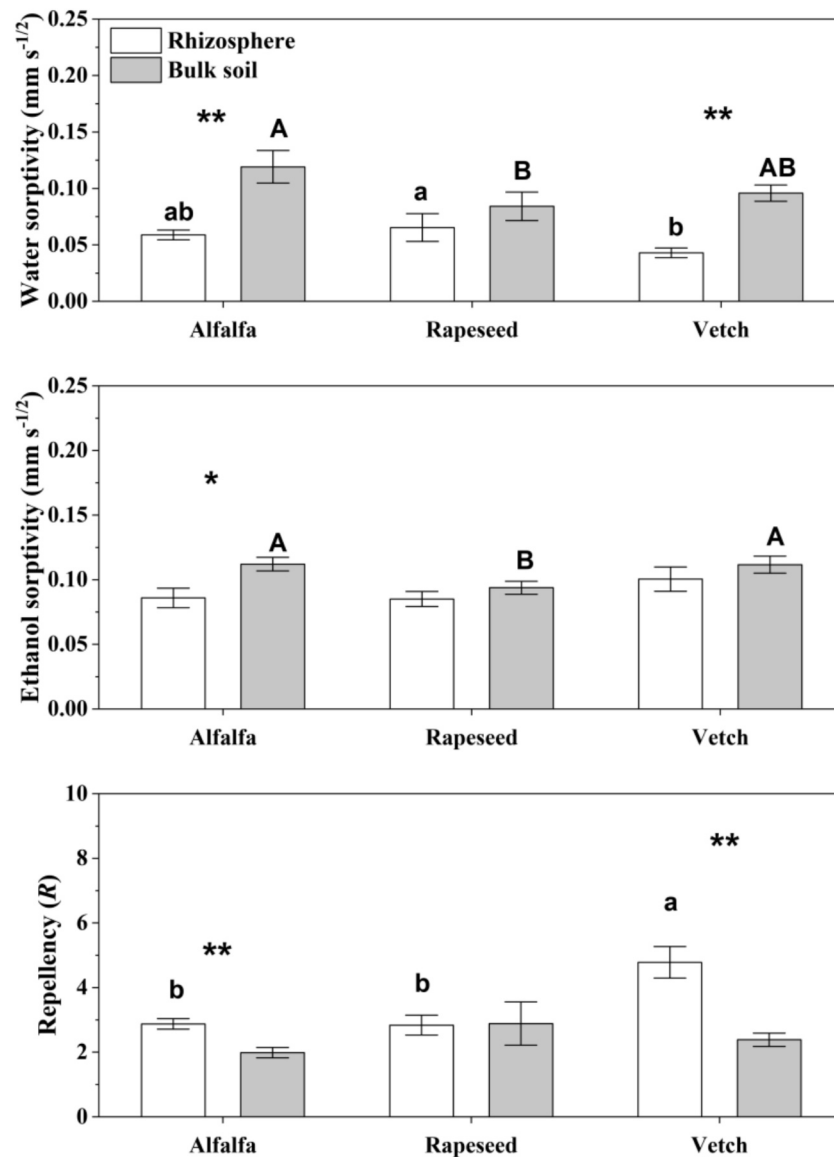


Fig. 6. Hydraulic properties of rhizosphere and bulk soil in three cover crop treatments. Different letters indicate significant differences among three cover crop treatments within the rhizosphere or bulk soil at $P < 0.05$; Stars indicate significant difference between the rhizosphere and bulk soil in each cover crop treatment (*, $P < 0.05$; **, $P < 0.01$). Means and standard errors are presented ($n = 10$). Samples were collected at 10–20 cm depth.

organic clogging as a fundamental constraint for water sorptivity.

Contrary to expectations from preferential flow theory (Luo et al., 2010; Zhang et al., 2019), CT-derived macropores showed no hydraulic connectivity in our sorptivity assays (Fig. 8). The lack of relationship between CT-derived macroporosity and sorptivities arises from a fundamental scale mismatch: while CT imaging quantified macropores $>12 \mu\text{m}$, the 10 mm head in our sorptivity assay restricts water entry to pores $<6 \mu\text{m}$ under unsaturated flow (Naveed et al., 2017). While macropores govern saturated hydraulic conductivity, their hydraulic relevance diminishes under unsaturated conditions, where water films preferentially occupy micropores (Mayes et al., 2003).

The strong matrix density-ethanol sorptivity relationship demonstrates that micropores serve as a critical determinant of sorptivity (Fig. 8), which is also supported by previous studies (Aravena et al., 2013; Hallett et al., 2003). However, the organic matter dominance introduces a partial regulatory effect: root-derived C inputs simultaneously reduce matrix density (Fig. S5) and promote micropores clogging, implying a strong interaction between matrix density and organic matter in controlling sorptivities. When controlling for total C effects,

matrix density independently explained 20.7% of ethanol sorptivity variation (Fig. 8), highlighting its fundamental yet non-negligible influence. Such regulation underscores the necessity of integrated matrix density-organic matter models for predicting rhizosphere hydraulics, particularly in cover crop systems where rapid root growth drives dynamic feedbacks between pore structure and carbon cycling (Hallett et al., 2003; Sun et al., 2022).

4.4. Implications for plant resilience, soil water availability, and future research

Rapeseed, by fostering a high-macroporosity, low-matrix density rhizosphere with minimal hydrophobicity (Figs. 3, 4, and 6), facilitated rapid rhizosphere-water infiltration and root expansion, particularly in the high-density soil (Rabbi et al., 2018). Conversely, the significantly enhanced water repellency in the alfalfa and vetch rhizosphere could indirectly promote plant drought resistance by slowing the rapid drying of the rhizosphere following wetting, potentially maintaining a more stable water potential within roots for prolonged periods during soil

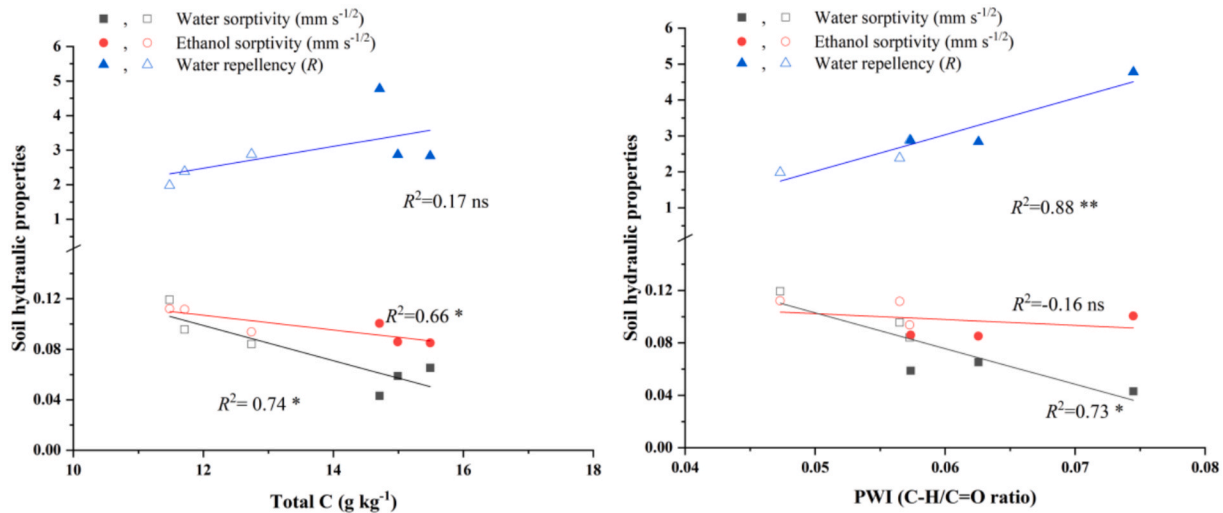


Fig. 7. Linear regression analysis ($n = 6$) between hydraulic properties and soil organic matter composition. Ns, not significant; *, $P < 0.05$; **, $P < 0.01$. Solid and hollow symbols represent the values of rhizosphere and bulk soil, respectively.

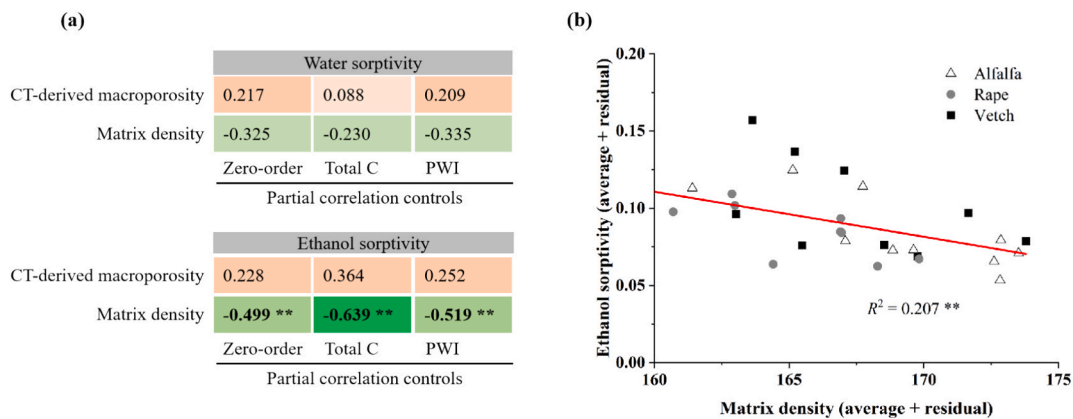


Fig. 8. Partial correlation and regression analyses ($n = 60$) between rhizosphere hydraulic properties and CT-derived soil characteristics. **, $P < 0.01$. (a) The correlation between rhizosphere hydraulic properties (water and ethanol sorptivities) and CT-derived soil characteristics (CT-derived macroporosity and matrix density), controlled for total C or PWI. Difference between zero-order and each partial correlation indicates the level of dependency of the correlation between controlled variable and water and ethanol sorptivities. The color and numbers indicate the sign (yellow and green colors indicate positive and negative correlations, respectively) and strength (the intensity of the color) of the correlation. (b) The scattergram plot and regression analysis between rhizosphere matrix density and ethanol sorptivity. The data were obtained from the partial residuals between rhizosphere matrix density and total C, as well as from the partial residuals between rhizosphere ethanol sorptivity and total C. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

drying (Kroener et al., 2016; Kruthika et al., 2024). Root growth preference and water acquisition strategies explain these adaptive paradoxes of cover crops (Geers-Lucas et al., 2025): the utilizing pre-existing macropores for rapeseed versus strong biological drilling capacity and organic-matter input capacity for alfalfa and vetch.

Notwithstanding these insights, our study has a limitation that warrant consideration and guide future research. By focusing exclusively on taproots to isolate their biomechanical and biochemical effects, we do not account for the hydraulic role of lateral roots, which are primary locations of water uptake. Nevertheless, the pore networks and organic matter gradients induced by taproots provide a critical foundation for understanding system-level hydraulics. Future work should integrate lateral roots to fully resolve species-specific hydraulic strategies across the entire root system.

5. Conclusion

Our field-based investigation demonstrates cover crop-specific rhizosphere engineering capacities in hydraulic properties for a

Vertisol. The rapeseed roots preferentially enhanced CT-derived macroporosity probably through using pre-existing macropores, whereas the alfalfa and vetch roots selectively increased rhizosphere total C and PWI via enriched hydrophobic groups. Contrary to conventional root compaction paradigms, no soil densification in the rhizosphere was observed likely due to exploiting pre-existing macropores and rhizosphere loosening through organic matter-mediated matrix density reduction. Meanwhile, a greater accumulation of hydrophobic aliphatic-C in the alfalfa and vetch rhizosphere, resulted in a greater PWI. Rhizosphere hydraulic functionality resulted from matrix density-organic matter interplay: while matrix density (rather than CT-derived pores) was associated with rhizosphere ethanol sorptivity variation, organic matter composition dominated rhizosphere water sorptivity through PWI-mediated water repellency and organic matter-induced pore clogging. These findings redefine rhizosphere hydraulic properties by establishing organic matter its content and composition as the primary determinants, with pore structure playing a critical role.

CRedit authorship contribution statement

Renjie Ruan: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Wanyu Xie:** Data curation. **Zhongbin Zhang:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Paul Hallett:** Writing – review & editing, Validation, Supervision. **Peng Xiong:** Funding acquisition, Data curation. **Shuai Liu:** Writing – review & editing, Formal analysis, Data curation. **Xinhua Peng:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was supported by Basic Research Center, Innovation Program of Chinese Academy of Agricultural Sciences (CAAS-BRC-AL-2025-02), Strategic Priority Research Program of the Chinese Academy of Sciences (XDA28010401), National Key Research and Development Program of China (2023YFD1501800), Jiangxi Provincial Science and Technology Special Project of Jinggangshan Agricultural High-tech Zone (20222-051261-2-2), and Project of Science and Technology Plan by Education Department of Jiangxi Province (GJJ2200445).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2026.135018>.

Data availability

Data will be made available on request.

References

- Angst, G., Kögel-Knabner, I., Kirfel, K., Hertel, D., Mueller, C.W., 2016. Spatial distribution and chemical composition of soil organic matter fractions in rhizosphere and non-rhizosphere soil under European beech (*Fagus sylvatica* L.). *Geoderma* 264, 179–187. <https://doi.org/10.1016/j.geoderma.2015.10.016>.
- Aravena, J.E., Berli, M., Ghezzehei, T.A., Tyler, S.W., 2011. Effects of root-induced compaction on rhizosphere hydraulic properties - X-ray microtomography imaging and numerical simulations. *Environ. Sci. Technol.* 45 (2), 425–431. <https://doi.org/10.1021/es102566j>.
- Aravena, J.E., et al., 2013. Synchrotron X-ray microtomography -new means to quantify root induced changes of rhizosphere physical properties. In: Anderson, S.H., Hopmans, J.W. (Eds.), *Soil-Water-Root Processes: Advances in Tomography and Imaging*. SSSA Special Publications, pp. 39–67. DOI:10.2136/sssaspecpub61.c3.
- Athmann, M., Sondermann, J., Kautz, T., Köpke, U., 2019. Comparing macropore exploration by faba bean, wheat, barley and oilseed rape roots using in situ endoscopy. *J. Soil Sci. Plant Nutr.* 19 (3), 689–700. <https://doi.org/10.1007/s42729-019-00069-0>.
- Atkinson, J.A., Hawkesford, M.J., Whalley, W.R., Zhou, H., Mooney, S.J., 2020. Soil strength influences wheat root interactions with soil macropores. *Plant, Cell Environ.* 43 (1), 235–245. <https://doi.org/10.1111/pce.13659>.
- Bengough, A., 2012. Water dynamics of the root zone: rhizosphere biophysics and its control on soil hydrology. *Vadose Zone J.* 11 (2), vjz2011. <https://doi.org/10.2136/vjz2011.0111>.
- Berli, M., Carminati, A., Ghezzehei, T.A., Or, D., 2008. Evolution of unsaturated hydraulic conductivity of aggregated soils due to compressive forces. *Water Resour. Res.* 44 (5). <https://doi.org/10.1029/2007WR006501>.
- Blanco-Canqui, H., et al., 2009. No-till induced increase in organic carbon reduces maximum bulk density of soils. *Soil Sci. Soc. Am. J.* 73 (6), 1871–1879. <https://doi.org/10.2136/sssaj2008.0353>.
- Blanco-Canqui, H., Ruis, S., 2020. Cover crop impacts on soil physical properties: a review. *Soil Sci. Soc. Am. J.* 84. <https://doi.org/10.1002/saj2.20129>.
- Burr-Hersey, J.E., Ritz, K., Bengough, G.A., Mooney, S.J., 2020. Reorganisation of rhizosphere soil pore structure by wild plant species in compacted soils. *J. Exp. Bot.* 71 (19), 6107–6115. <https://doi.org/10.1093/jxb/era323>.
- Carminati, A., et al., 2013. Do roots mind the gap? *Plant and Soil* 367 (1–2), 651–661. <https://doi.org/10.1007/s11104-012-1496-9>.
- Carminati, A., Zarebanadkouki, M., Kroener, E., Ahmed, M.A., Holz, M., 2016. Biophysical rhizosphere processes affecting root water uptake. *Ann. Bot.* 118 (4), 561–571. <https://doi.org/10.1093/aob/mcw113>.
- Chen, G., Weil, R.R., 2010. Penetration of cover crop roots through compacted soils. *Plant and Soil* 331 (1), 31–43. <https://doi.org/10.1007/s11104-009-0223-7>.
- Daly, K.R., et al., 2015. Assessing the influence of the rhizosphere on soil hydraulic properties using X-ray computed tomography and numerical modelling. *J. Exp. Bot.* 66 (8), 2305–2314. <https://doi.org/10.1093/jxb/eru509>.
- Dexter, A.R., 1987. Compression of soil around roots. *Plant and Soil* 97 (3), 401–406. <https://doi.org/10.1007/bf02383230>.
- Du, H., Yu, G., Guo, M., Xu, H., 2021. Investigation of carbon dynamics in rhizosphere by synchrotron radiation-based Fourier transform infrared combined with two dimensional correlation spectroscopy. *Sci. Total Environ.* 762, 143078. <https://doi.org/10.1016/j.scitotenv.2020.143078>.
- Duddek, P., et al., 2023. The effect of root hairs on root water uptake is determined by root-soil contact and root hair shrinkage. *New Phytol.* 240 (6), 2484–2497. <https://doi.org/10.1111/nph.19144>.
- Geers-Lucas, M., Guber, A., Kravchenko, A., 2025. Root-pore interactions, the underestimated driver for rhizosphere structure and rhizosheath development. *Plant Cell Environ.* <https://doi.org/10.1111/pce.70215>.
- Gregory, P.J., 2006. Roots, rhizosphere and soil: the route to a better understanding of soil science? *Eur. J. Soil Sci.* 57 (1), 2–12. <https://doi.org/10.1111/j.1365-2389.2005.00778.x>.
- Hallett, P.D., Gordon, D.C., Bengough, A.G., 2003. Plant influence on rhizosphere hydraulic properties: direct measurements using a miniaturized infiltrometer. *New Phytol.* 157 (3), 597–603. <https://doi.org/10.1046/j.1469-8137.2003.00690.x>.
- Heller, C., Ellerbrock, R.H., Roßkopf, N., Kligenfuß, C., Zeitz, J., 2015. Soil organic matter characterization of temperate peatland soil with FTIR-spectroscopy: effects of mire type and drainage intensity. *Eur. J. Soil Sci.* 66 (5), 847–858. <https://doi.org/10.1111/ejss.12279>.
- Helliwell, J.R., et al., 2017. The emergent rhizosphere: imaging the development of the porous architecture at the root-soil interface. *Sci. Rep.* 7 (1), 14875. <https://doi.org/10.1038/s41598-017-14904-w>.
- Helliwell, J.R., Sturrock, C.J., Miller, A.J., Whalley, W.R., Mooney, S.J., 2019. The role of plant species and soil condition in the structural development of the rhizosphere. *Plant Cell Environ.* 42 (6), 1974–1986. <https://doi.org/10.1111/pce.13529>.
- Huang, G., et al., 2022. Ethylene inhibits rice root elongation in compacted soil via ABA- and auxin-mediated mechanisms. *PNAS* 119 (30), e2201072119. <https://doi.org/10.1073/pnas.2201072119>.
- Jordán, A., Zavala, L.M., Mataix-Solera, J., Doerr, S.H., 2013. Soil water repellency: origin, assessment and geomorphological consequences. *Catena* 108, 1–5. <https://doi.org/10.1016/j.catena.2013.05.005>.
- Katuwal, S., Moldrup, P., Lamandé, M., Tuller, M., de Jonge, L.W., 2015. Effects of CT number derived matrix density on preferential flow and transport in a macroporous agricultural soil. *Vadose Zone J.* 14 (7), vjz2015. <https://doi.org/10.2136/vjz2015.01.0002>.
- Koebnick, N., et al., 2019. Imaging microstructure of the barley rhizosphere: particle packing and root hair influences. *New Phytol.* 221 (4), 1878–1889. <https://doi.org/10.1111/nph.15516>.
- Koebnick, N., et al., 2017. High-resolution synchrotron imaging shows that root hairs influence rhizosphere soil structure formation. *New Phytol.* 216 (1), 124–135. <https://doi.org/10.1111/nph.14705>.
- Koebnick, N., Schluter, S., Blaser, S., Vetterlein, D., 2018. Root-soil contact dynamics of *Vicia faba* in sand. *Plant and Soil* 431 (1–2), 417–431. <https://doi.org/10.1007/s11104-018-3769-4>.
- Kroener, E., Zarebanadkouki, M., Bittelli, M., Carminati, A., 2016. Simulation of root water uptake under consideration of nonequilibrium dynamics in the rhizosphere. *Water Resour. Res.* 52 (8), 5755–5770. <https://doi.org/10.1002/2015WR018579>.
- Kruthika, S., et al., 2024. Unveiling the role of root exudates in plant adaptation to drought and heat stress. *J. Crop Health* 76 (5), 941–955. <https://doi.org/10.1007/s10343-024-01013-8>.
- Kunlanit, B., Vityakon, P., Puttaso, A., Cadisch, G., Rasche, F., 2014. Mechanisms controlling soil organic carbon composition pertaining to microbial decomposition of biochemically contrasting organic residues: evidence from midDRIFTS peak area analysis. *Soil Biol. Biochem.* 76, 100–108. <https://doi.org/10.1016/j.soilbio.2014.05.006>.
- Leue, M., Ellerbrock, R.H., Gerke, H.H., 2010. DRIFT mapping of organic matter composition at intact soil aggregate surfaces. *Vadose Zone J.* 9 (2), 317–324. <https://doi.org/10.2136/vjz2009.0101>.
- Leue, M., Gerke, H.H., Ellerbrock, R.H., 2013. Millimetre-scale distribution of organic matter composition at intact biopore and crack surfaces. *Eur. J. Soil Sci.* 64 (6), 757–769. <https://doi.org/10.1111/ejss.12098>.
- Leue, M., Gerke, H.H., Godow, S.C., 2015. Droplet infiltration and organic matter composition of intact crack and biopore surfaces from clay-illuvial horizons. *J. Plant Nutr. Soil Sci.* 178 (2), 250–260. <https://doi.org/10.1002/jpln.201400209>.
- Lin, L., et al., 2022. Influence of soil physical and chemical properties on mechanical characteristics under different cultivation durations with Mollisols. *Soil Tillage Res.* 224, 105520. <https://doi.org/10.1016/j.still.2022.105520>.
- Lucas, M., Nguyen, L.T.T., Guber, A., Kravchenko, A.N., 2022. Cover crop influence on pore size distribution and biopore dynamics: enumerating root and soil faunal effects. *Front. Plant Sci.* 13–2022. <https://doi.org/10.3389/fpls.2022.928569>.
- Lucas, M., Santiago, J.P., Chen, J., Guber, A., Kravchenko, A., 2023. The soil pore structure encountered by roots affects plant-derived carbon inputs and fate. *New Phytol.* 240 (2), 515–528. <https://doi.org/10.1111/nph.19159>.

- Luo, L., Lin, H., Li, S., 2010. Quantification of 3-D soil macropore networks in different soil types and land uses using computed tomography. *J. Hydrol.* 393 (1), 53–64. <https://doi.org/10.1016/j.jhydrol.2010.03.031>.
- Mayes, M.A., et al., 2003. Transport of multiple tracers in variably saturated humid region structured soils and semi-arid region laminated sediments. *J. Hydrol.* 275 (3), 141–161. [https://doi.org/10.1016/S0022-1694\(03\)00039-8](https://doi.org/10.1016/S0022-1694(03)00039-8).
- Mikutta, C., Lang, F., Kaupenjohann, M., 2004. Soil organic matter clogs mineral pores. *Soil Sci. Soc. Am. J.* 68 (6), 1853–1862. <https://doi.org/10.2136/sssaj2004.1853>.
- Naveed, M., et al., 2017. Rhizosphere-scale quantification of hydraulic and mechanical properties of soil impacted by root and seed exudates. *Vadose Zone J.* 17 (1), 1–12. <https://doi.org/10.2136/vzj2017.04.0083>.
- Paradelo, M., et al., 2016. X-ray CT-derived soil characteristics explain varying air, water, and solute transport properties across a loamy field. *Vadose Zone J.* 15 (4), vzj2015. <https://doi.org/10.2136/vzj2015.07.0104>.
- Rabbi, S.M.F., et al., 2018. Plant roots redesign the rhizosphere to alter the three-dimensional physical architecture and water dynamics. *New Phytol.* 219 (2), 542–550. <https://doi.org/10.1111/nph.15213>.
- Rumpel, C., Kögel-Knabner, I., 2011. Deep soil organic matter—a key but poorly understood component of terrestrial C cycle. *Plant and Soil* 338 (1), 143–158. <https://doi.org/10.1007/s11104-010-0391-5>.
- Song, X., Fang, C., Yuan, Z.-Q., Li, F.-M., 2021. Long-term growth of alfalfa increased soil organic matter accumulation and nutrient mineralization in a semi-arid environment. *Front. Environ. Sci.* 9–2021. <https://doi.org/10.3389/fenvs.2021.649346>.
- Soil Survey Staff, 2015. Illustrated guide to soil taxonomy. U.S. Department of Agriculture, Natural Resources Conservation Service, National Soil Survey Center, Lincoln, Nebraska.
- Sun, F., Xiao, B., Kidron, G.J., 2022. Towards the influences of three types of biocrusts on soil water in drylands: insights from horizontal infiltration and soil water retention. *Geoderma* 428, 116136. <https://doi.org/10.1016/j.geoderma.2022.116136>.
- Verchot, L.V., Dutaur, L., Shepherd, K.D., Albrecht, A., 2011. Organic matter stabilization in soil aggregates: understanding the biogeochemical mechanisms that determine the fate of carbon inputs in soils. *Geoderma* 161 (3–4), 182–193. <https://doi.org/10.1016/j.geoderma.2010.12.017>.
- Villarreal, R., et al., 2024. Hydraulic characterization of earthworm macropore surfaces using miniaturized infiltration data. *Soil Tillage Res.* 236. <https://doi.org/10.1016/j.still.2023.105954>.
- Whalley, W.R., et al., 2005. Structural differences between bulk and rhizosphere soil. *Eur. J. Soil Sci.* 56 (3), 353–360. <https://doi.org/10.1111/j.1365-2389.2004.00670.x>.
- Xiong, P., Zhang, Z., Wang, Y., Peng, X., 2022. Variable responses of maize roots at the seedling stage to artificial biopores in noncompacted and compacted soil. *J. Soil. Sediment.* 22 (4), 1155–1164. <https://doi.org/10.1007/s11368-021-03133-4>.
- Zarebanadkouki, M., Kim, Y.X., Carminati, A., 2013. Where do roots take up water? Neutron radiography of water flow into the roots of transpiring plants growing in soil. *New Phytol.* 199 (4), 1034–1044. <https://doi.org/10.1111/nph.12330>.
- Zhang, Z., et al., 2018. Three dimensional characteristics of biopores and non-biopores in the subsoil respond differently to land use and fertilization. *Plant and Soil* 428 (1), 453–467. <https://doi.org/10.1007/s11104-018-3689-3>.
- Zhang, Z., et al., 2019. Linking saturated hydraulic conductivity and air permeability to the characteristics of biopores derived from X-ray computed tomography. *J. Hydrol.* 571, 1–10. <https://doi.org/10.1016/j.jhydrol.2019.01.041>.
- Zhang, Z., Peng, X., 2021. Bio-tillage: a new perspective for sustainable agriculture. *Soil Tillage Res.* 206, 104844. <https://doi.org/10.1016/j.still.2020.104844>.
- Zhang, Z., et al., 2022. Bio-tillage improves soil physical properties and maize growth in a compacted Vertisol by cover crops. *Soil Sci. Soc. Am. J.* 86 (2), 324–337. <https://doi.org/10.1002/saj2.20368>.
- Zheng, C., et al., 2023. Plant root plasticity during drought and recovery: what do we know and where to go? *Front. Plant Sci.* 14. <https://doi.org/10.3389/fpls.2023.1084355>.